

Ilyes Gharbi

Electrical Engineer

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SUMMARY

Embedded Systems Engineer with experience in the automotive wiring harness industry, passionate about industrial processes, automation, and smart manufacturing. Familiar with cable harness production environments, process improvement methodologies, and problem-solving approaches. Curious, rigorous, and solution-oriented, I aim to apply my technical skills and analytical mindset to optimize manufacturing processes and contribute to high-quality automotive projects.

PROFESSIONAL EXPERIENCE

Final Year Project Intern – SEBN, Jendouba

02/2025 – 06/2025

- Designed and implemented an autonomous robot based on ROS2, LiDAR, and embedded camera.
- Developed and controlled a robotic arm for product handling and storage.
- Built and integrated a mobile application (React Native) for real-time supervision, traceability, and inventory management.

Engineering Intern – SEBN, Jendouba

07/2024 – 08/2024

EDUCATION

Engineering Degree in Electrical Engineering

2022 – 2025

International Multidisciplinary School of Sousse

Bachelor's Degree in Electrical Engineering

2018 – 2022

Higher Institute of Technological Studies of Jendouba

SKILLS

Programming :

C, C++, Python, Ladder, VHDL, MATLAB/Simulink

electronics

Arduino, Raspberry Pi, ESP8266, FPGA, STM32

Design Software

Proteus, PSIM, Modelsim, LABVIEW

IOT platforms

Blynk, Arduino IOT cloud, Node-RED

Communication Protocols

UART, SPI, I2C, CAN, MQTT

Operating Systems

Windows, Linux, FreeRTOS

CERTIFICATES

- Node-RED Fundamentals
- Six Sigma Yellow Belt [🔗](#)

LANGUAGES

- Arabic - Native
- English - autonomous (B1)
- French - autonomous (B1)
- German - basic (A1)

PROJECTS

Steg Gas Pressure Regulating Station

2026

Developed an IoT-based monitoring system for a STEG gas pressure regulating station, enabling real-time monitoring of gas pressure, temperature, and safety parameters through a SCADA dashboard.

key words : ESP32, Arduino IDE, MQTT, Mosquitto, Node-RED, SCADA, IOT

Smart Factory Environmental Monitoring System

2026

Developed an Industry 4.0 IoT-based environmental monitoring system for industrial applications, enabling real-time monitoring of temperature, humidity, air quality, and safety parameters through a SCADA dashboard. Implemented ESP32-based data acquisition, MQTT communication via Mosquitto, and Node-RED visualization for real-time supervision and alarm management.

Keywords: ESP32, Arduino IDE, MQTT, Mosquitto, Node-RED, SCADA, IoT, Industry 4.0.